Jahid Hasan

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Research Summary

Modern data scientists devote significant effort to data-intensive tasks such as data discovery, cleaning, pipeline optimization, and debugging. My research rethinks this paradigm by developing intelligent, data-centric systems that automate these processes while ensuring reliability, robustness, and adaptability under deployment-time adversities. At the intersection of machine learning, database systems, and agentic AI, I focus on automated ML pipeline diagnosis, data preparation, and adaptive interventions to build trustworthy and efficient ML systems. My research has resulted in first-author publications at leading data management venues, including **VLDB**, **ICDE**, and **EDBT**.

Research Interests: *Data-Centric AI · Agentic AI · Foundation Models · Data Management · Data Quality · Trustworthy and Responsible AI · ML Systems · Data Shift · Pipeline Management · Information Retrieval*

Education

Ph.D. in Computer and Information Technology August 2023 – May 2027 (Anticipated)
Purdue University, West Lafayette, IN, USA CGPA: 3.94/4.00
Dissertation: *Intelligent Data-Centric Systems for Fair, Robust, and Adaptive Data Science Pipelines*
Supervisor: Dr. Romila Pradhan

M.S. in Mathematics January 2022 – July 2023
University of Dhaka, Dhaka, Bangladesh CGPA: 3.81/4.00
Dissertation: *Age Distributed Model of Communicable Disease in Heterogeneous Environment*

B.S. in Mathematics January 2017 – December 2021
University of Dhaka, Dhaka, Bangladesh CGPA: 3.69/4.00

Research Experience

Graduate Research Assistant Aug 2023 – Present
Responsible Data Science Lab, Purdue University, USA

Leading technical efforts in the NSF (Award Number: 2237149)- and CASMI-funded projects *"Data Preparation for Fair and Trustworthy Machine Learning"*; additionally collaborating with AT&T Research Lab, Cornell University, Université Paris Cité, France, Purdue Virtual Reality Lab, and Purdue Computer Science.

Selected Research Projects:

- Benchmarking Multi-Agent Systems for Data Quality Debugging and Repair — Ongoing
- Agentic AI Data Fitness: Task-Aware Assessment and Adaptive Data Interventions — Targeting@ VLDB 2027
- PipeLens: Identifying Interventions for Resolving Malfunctioning Data Science Pipelines [1] — VLDB 2026
- DataSift: Selective Data Expansion for ML Model Performance [2] — EDBT 2026
- Sentinel: Evaluating Pipeline Robustness Under Distributional Shifts [3, 4] — ICDE 2026
- FanShield: Attention-Enhanced Intelligent Obstacle Avoidance in Extended Reality — Under Review@ CHI 2027

Graduate Research Assistant Jan 2022 – Jul 2023
University of Dhaka, Dhaka, Bangladesh

- Age Distributed Model of Communicable Disease in Heterogeneous Environment

Selected Publications

[1] Jahid Hasan, Stanley Jiang, Tejendra Singh, Sainyam Galhotra, Romila Pradhan, and Divesh Srivastava. PipeLens: Identifying interventions for resolving malfunctioning data science pipelines. *Proceedings of the VLDB Endowment*, 19, 2026. 52nd International Conference on Very Large Data Bases (VLDB 2026).

[2] Jahid Hasan and Romila Pradhan. Selective data expansion for model performance. In *Proceedings of the International Conference on Extending Database Technology (EDBT)*, Tampere, Finland, 2026. OpenProceedings.

[3] Jahid Hasan, Jingya Wang, and Romila Pradhan. SENTINEL: Evaluating Pipeline Robustness to Distributional Shifts. In *Proceedings of the 42nd IEEE International Conference on Data Engineering (ICDE)*, 2026.

[4] Jahid Hasan and Romila Pradhan. Explanations for machine learning pipelines under data drift. In *Proceedings of the Workshop on Human-In-the-Loop Data Analytics*, HILDA '25, New York, NY, USA, 2025. Association for Computing Machinery.

Teaching Experience

Teaching Assistant Spring 2026 & Spring 2027

Computer and Information Technology, Purdue University CNIT 39200: Enterprise Data Management

· Instructed four sections of Database Lab as an alternate Teaching Assistant over the semester.

Research Mentorship

Ritwik Jayaraman, Computer Science, Purdue University 2026-2027
As part of the 2026 NSF CISE REU Program.
Project Title: *"Benchmarking Multi-Agent System on Data Quality Assesment".*

Wail Ouaret, National School of Artificial Intelligence, Algeria 2026
As part of the collaboration with the Université Paris Cité. (Manuscript Under Review @ VLDB 2027)
Project Title: *"ReFair: Regional Performance Diagnosis and Selective Model Adaptation under Data Drift".*

Burton Lu, Computer and Information Technology, Purdue University 2025-2026
Project Title: *"Evaluating Performance-Based Drift Detection for Localizing Data Corruption in ML Pipelines".*

Technical Skills

Programming	Python, MATLAB, SQL, R, Fortran, SAS, SPSS
Machine Learning	scikit-learn, PyTorch, TensorFlow, Hugging Face, LLMs, NLP
IR & Agentic AI	RAG, LangChain, Embeddings, Vector Database
Data & Systems	Pandas, NumPy, MySQL, MongoDB
Visualization	Matplotlib, Seaborn, Tableau

Scholarly Service

PC Member, DQ4AI @ EDBT 2027
Shadow PC Member, VLDB 2027
Reviewer for Elsevier journals, TPAMI, ACM FAccT, PST 2025-Present
External Reviewer for ICDE, SIGMOD, VLDB, EDBT, and CIKM 2024-Present
Judge, Undergraduate Research Symposium, Purdue University 2025, 2026

Licenses & Certifications

EDBT–Microsoft Summer School on AI and Data Management July 2025
EDBT Association, Nicosia, Cyprus

Responsible Conduct of Research (RCR) Training Feb 2024 – Feb 2028
CITI Program Credential ID: 61203291

Social & Behavioral Research Investigators Certification Feb 2024 – Feb 2028
CITI Program Credential ID: 61203290

Advanced Learning Algorithms 2024
DeepLearning.AI, Stanford University

Unsupervised Learning, Recommenders, Reinforcement Learning 2024
DeepLearning.AI, Stanford University

Supervised Machine Learning: Regression and Classification Feb 2023
DeepLearning.AI, Stanford University Credential ID: KNJ76GCJGYHV

Machine Learning with Python: Foundations Nov 2023
LinkedIn Learning

Conference Participation & Academic Outreach

The 52nd International Conference on Very Large Database (VLDB), Boston, MA, USA 2026

42nd IEEE International Conference on Data Engineering (ICDE), Montréal, Canada 2026

International Conference on Extending Database Technology (EDBT), Tampere, Finland 2026

ACM SIGMOD/PODS International Conference on Management of Data, Berlin, Germany 2025

EDBT Summer School on AI and Data Management, Nicosia, Cyprus 2025

Workshop on Digital Safety, UL Research Institute & Northwestern University, Chicago 2024

Professional Affiliations

Association for Computing Machinery (ACM)

IEEE Computational Intelligence Society (IEEE CIS)

ACM SIGMOD

Phi Kappa Phi Honor Society

Awards and Grants

Travel Grant from Applied AI Research Center 2026

Travel Grant from Purdue Graduate Student Government (PGSG) 2026

Microsoft EDBT US Fellowship 2025
Awarded for research excellence in data management for fair AI, which covered the travel expense to Cyprus.

Travel Grant, Digital Safety Workshop, UL Research Institute 2024

National Science and Technology (NST) Fellowship, Bangladesh 2022
Awarded by the Ministry of Science and Technology for outstanding academic performance and research potential.

Invited Talks

- **Guest Lecturer**, Computer Science, Purdue University-Northwest Campus Spring & Fall 2026
  "Can We Trust AI? Understanding Reliability, Fairness, and Bias"
- **Research Talk**, Applied AI Research Center, Purdue University Fall 2026
  "Data-Centric ML Pipeline Management: From Failure Diagnosis to Adaptive Intervention"

Leadership & Community Involvement

- **Senator**, Purdue Graduate Student Government, Purdue University 2026 – Present
  Elected to represent the Computer and Information Technology department in the Graduate Student Government, contributing to legislative decisions impacting a graduate student body of over 12,000 students.
- **Event Director**, Polytechnic Graduate Student Organization, Purdue University 2024 – 2025
  Co-founded this organization for 1,100+ techie graduate students at Purdue by drafting bylaws and building its structure, while leading events, budgets, and logistics that made it popular among students.
- **Fundraising Chair**, Bangladesh Student Association at Purdue 2024 – 2025
  Led fundraising strategy that secured $56,000 from Purdue's Student Fee Advisory Board (SFAB) to support cultural exchange activities, while organizing events and strengthening donor relationships.

References

Dr. Romila Pradhan
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Dr. Divesh Srivastava
Director
AT&T Research
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Dr. Sainyam Galhotra
Assistant Professor
Computer Science
Cornell University
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